

# CS-405 Discrete Mathematics

## Class Assignment

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Date: 05/09/2026

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      SUBROUTINE TRANS(A,N,FLAG)
C THIS SUBROUTINE IS INVOKED AFTER THE RELATION MATRIX A HAS BEEN
C PROVED TO REPRESENT A REFLEXIVE AND SYMMETRIC RELATION. IF THE
C RELATION IS TRANSITIVE, THEN FLAG, WHICH WAS RESET TO FALSE IS SET TO
C TRUE, INDICATING THAT THE RELATION IS AN EQUIVALENCE RELATION.
C OTHERWISE, FLAG REMAINS FALSE ON RETURN.
C
      LOGICAL*1 A(N,N),FLAG
C
C CHECK FOR TRIVIAL CASE: N = 1 OR N = 2
      IF(N.EQ.1.OR.N.EQ.2) GO TO 20
C
C I IS THE ROW COUNTER.
      NMIN2 = N - 2
      DO 10 I = 1,NMIN2
C
C J IS A COLUMN COUNTER RANGING FROM I+1 TO N-1.
      L = I + 1
      NMIN1 = N - 1
      DO 10 J = L,NMIN1
C
C FIND A PAIR (I,J) FOR WHICH I AND J ARE RELATED.
      IF(.NOT.A(I,J)) GO TO 10
C
C K SCANS FROM COLUMN J+1 TO N.
      M = J + 1
      DO 5 K = M,N
C
C FIND THE FOLLOWING RELATED PAIRS IN ROW I.
      IF(.NOT.A(I,K)) GO TO 5
C
C TRANSITIVE? A(I,J) AND A(I,K) IMPLIES A(K,J).
      IF(.NOT.A(K,J)) RETURN
      5 CONTINUE
      10 CONTINUE
C
C THE RELATION IS AN EQUIVALENCE RELATION.
      20 FLAG = .TRUE.
      RETURN
      END
```

FIGURE 2-3.10 FORTRAN subroutine for algorithm TRANS.

The given FORTRAN subroutine checks whether a relation on a set with **cardinality 'N'** is equivalent. It is already given that the relation, which is given by a **NxN matrix 'A'** is reflexive

and symmetric. The program just checks whether it is transitive. If  $A[x][y] = \text{True}$ , that indicates that the element at  $x^{\text{th}}$  position is related to the element at  $y^{\text{th}}$  position and  $A[x][y] = \text{False}$  indicates that they are not related.

The algorithm first checks for two trivial cases, where it automatically passes as a transitive relation if  $N \leq 2$ .

- **Case 1:  $N = 1$ .** Consider a set  $S = \{a\}$ . Here a reflexive and symmetric relation  $R$  on  $S$  can either be an empty set or it can be  $R = \{(a, a)\}$ . In both cases, by definition the relation is reflexive, symmetric and transitive.
- **Case 2:  $N = 2$ .** Consider a set  $S = \{a, b\}$ . Here a reflexive and symmetric relation  $R$  on  $S$  can be one of the following:
  - An empty set
  - $R = \{(a, a)\}$
  - $R = \{(b, b)\}$
  - $R = \{(a, a), (a, b), (b, a)\}$
  - $R = \{(b, b), (a, b), (b, a)\}$
  - $R = \{(a, a), (b, b)\}$
  - $R = \{(a, a), (b, b), (a, b), (b, a)\}$
- We can check that in all possible cases, the relation is also transitive.

The algorithm then iterates through all the rows of the matrix using a variable ' $i$ '. Within the main loop, it iterates through all the columns of the matrix using the variable ' $j$ '. Here ' $j$ ' does not need to iterate the entire length  $N$  of the matrix and it can just start at  $i + 1$ .

- We can skip all cases where  $i > j$ . This is because we know that the relation is symmetric. So if  $A[i][j]$  is True, then  $A[j][i]$  must also be True. And if  $A[i][j]$  is False, then  $A[j][i]$  must also be False.
- We can also skip all the cases where  $i == j$ . This is because we know that the relation is reflexive. So  $A[i][i]$  will be True for all values of  $i == j$ .

The algorithm now iterates through the remaining elements and checks if it finds a relation, i.e.,  $A[i][j] = \text{True}$ . If it does, then it iterates through the remaining elements in one last inner loop using the variable ' $k$ '.

- It checks whether  $A[j][k]$  is True. If it is True, it checks whether  $A[i][k]$  is also True.
- This is same as checking if  $(a, b) \in R$  and  $(b, c) \in R$ , then whether or not  $(a, c) \in R$
- If it  $A[i][k]$  turns out to be False, it sets the FLAG as False, indicating that the relation is not transitive, thus not an equivalence relation.
- If the algorithm passes all the steps, it sets the FLAG as True, indicating that the relation is transitive, thus is an equivalence relation.